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Ex.No:9 **Date:30/09/25**

DESIGN A SIMPLE TOPOLOGY USING CISCO PACKET TRACER

Aim:

To design and configure a simple network topology with one router, two switches, and four PCs in Cisco Packet Tracer for successful inter-network communication.

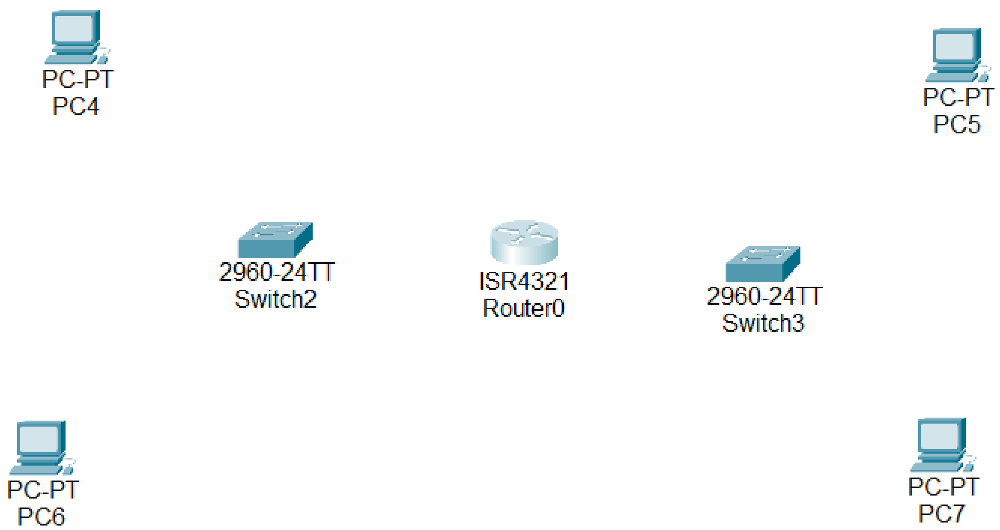
Procedure:

1. Launch Cisco Packet Tracer

- Open Cisco Packet Tracer on your system.
- Ensure the workspace and device categories (Routers, Switches, End Devices, etc.) are visible.

2. Add Network Devices

- From the Routers section, drag one router onto the workspace.
- From the Switches section, drag two 2960 switches.
- From the End Devices section, drag four PCs (PC1, PC2, PC3, PC4).
- Rename the devices.



3. Connect the Devices

Use Copper Straight-Through cables to connect:

- PC1 to Switch1 using port FastEthernet0/1.
- PC2 to Switch1 using port FastEthernet0/2.
- PC3 to Switch2 using port FastEthernet0/1.
- PC4 to Switch2 using port FastEthernet0/2.
- Switch1 to the router using Switch1 port GigabitEthernet0/1 and Router port GigabitEthernet0/0/0.
- Switch2 to the router using Switch2 port GigabitEthernet0/1 and Router port GigabitEthernet0/0/1.

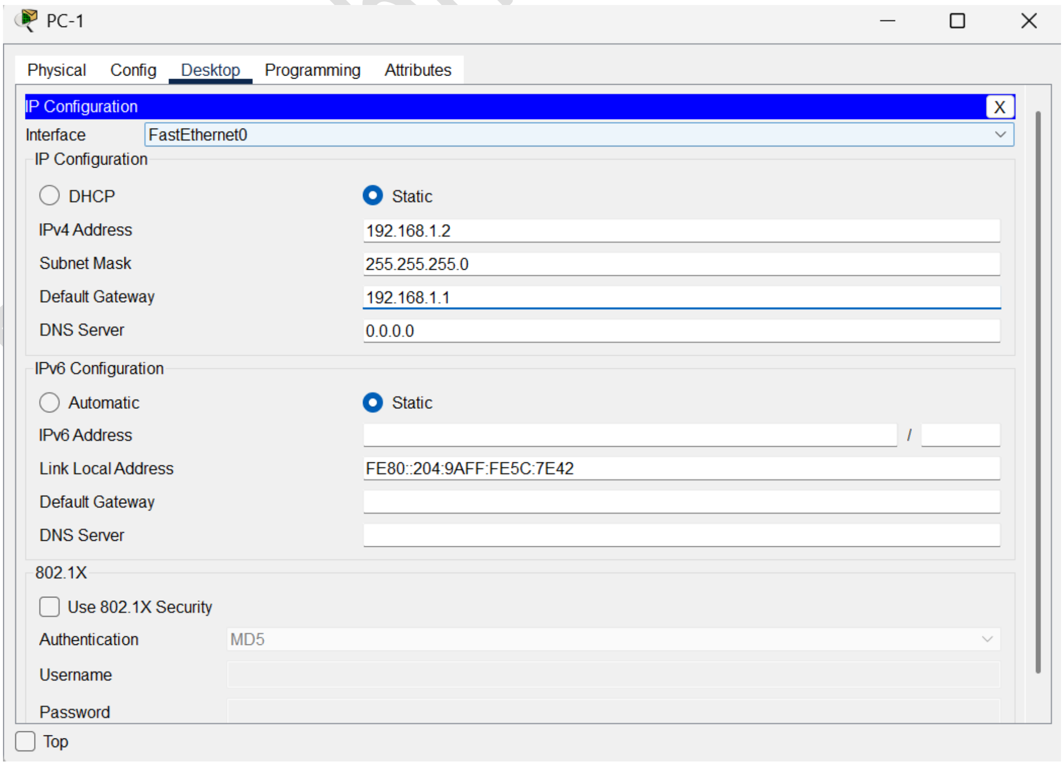
4. Assign IP Addresses:

- Network 1 (Switch1 side): 192.168.1.0/24
- Network 2 (Switch2 side): 192.168.2.0/24

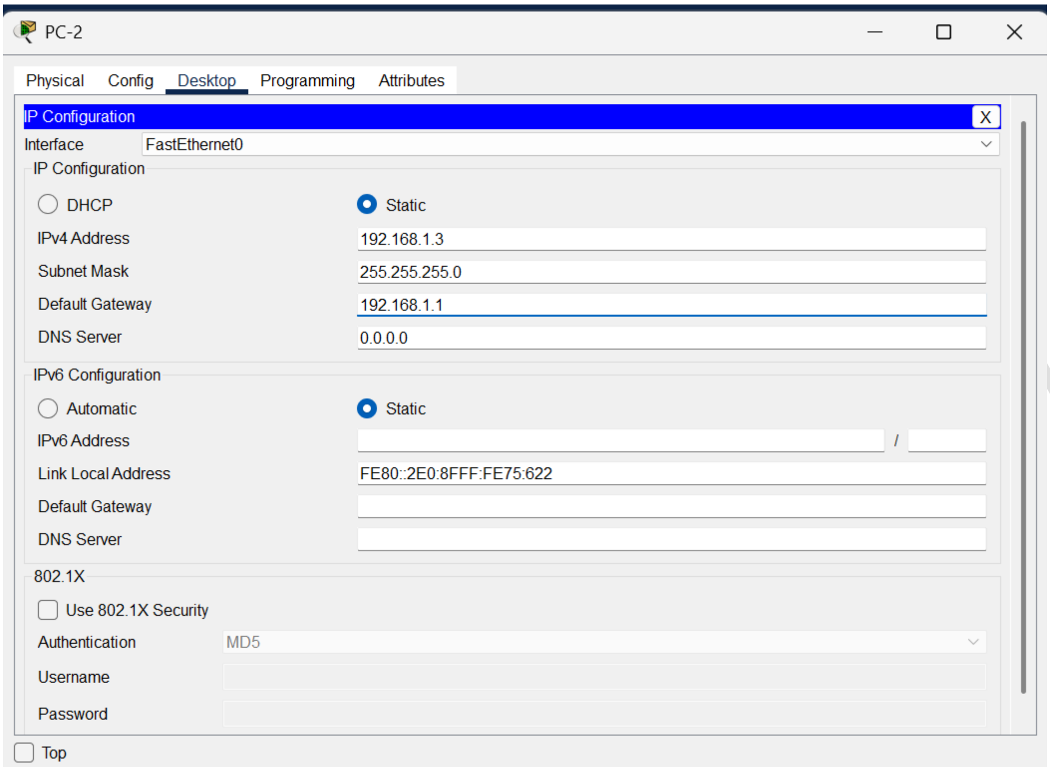
For each PC (PC1 to PC4):

1. Click the PC → go to Desktop → IP Configuration.
2. Select Static and manually enter the IP address, subnet mask, and default gateway

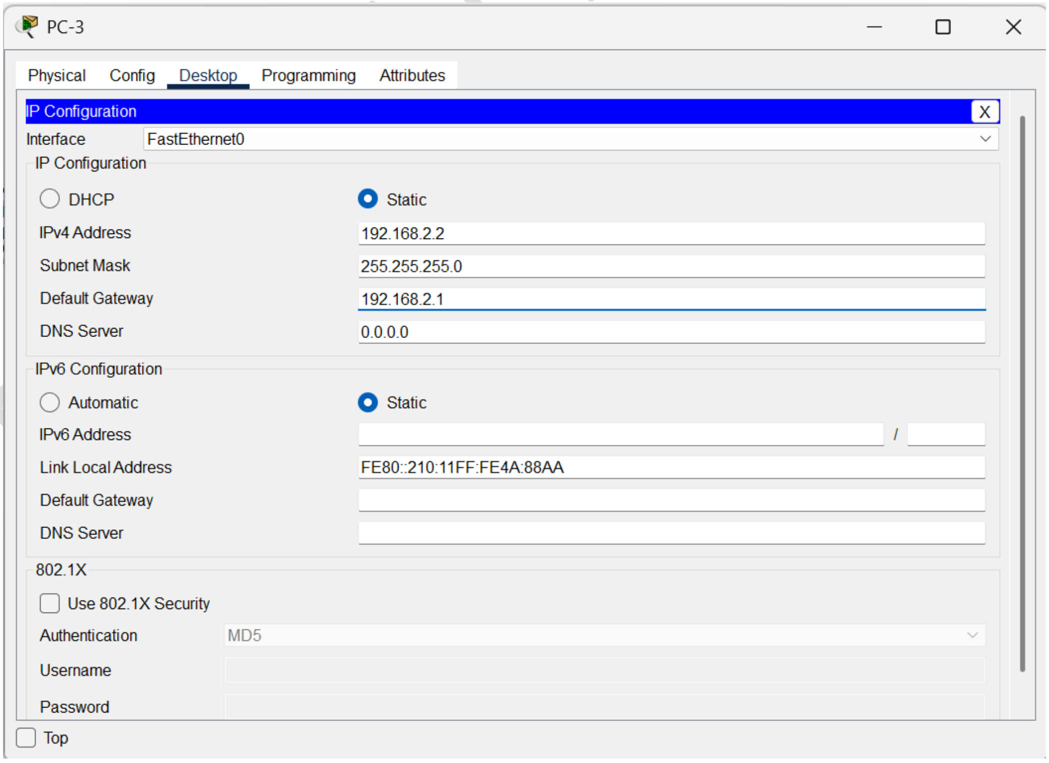
- PC1 → IP 192.168.1.2 / 255.255.255.0, Gateway 192.168.1.1



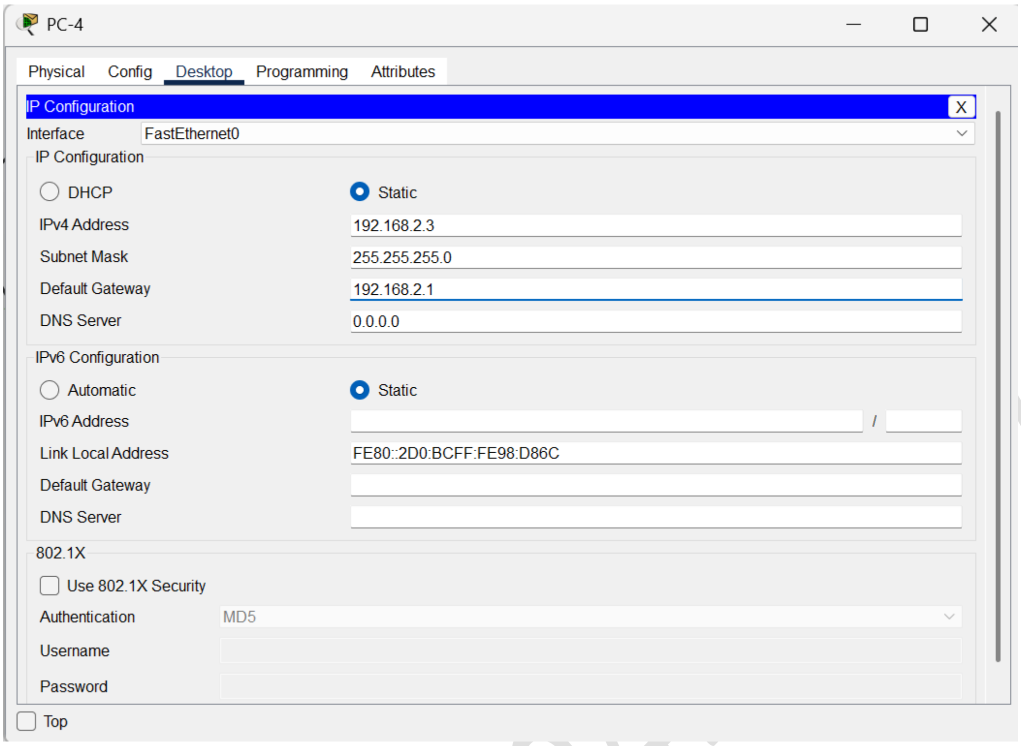
- PC2 → IP 192.168.1.3 / 255.255.255.0, Gateway 192.168.1.1



- PC3 → IP 192.168.2.2 / 255.255.255.0, Gateway 192.168.2.1



- PC4 → IP 192.168.2.3 / 255.255.255.0, Gateway 192.168.2.1

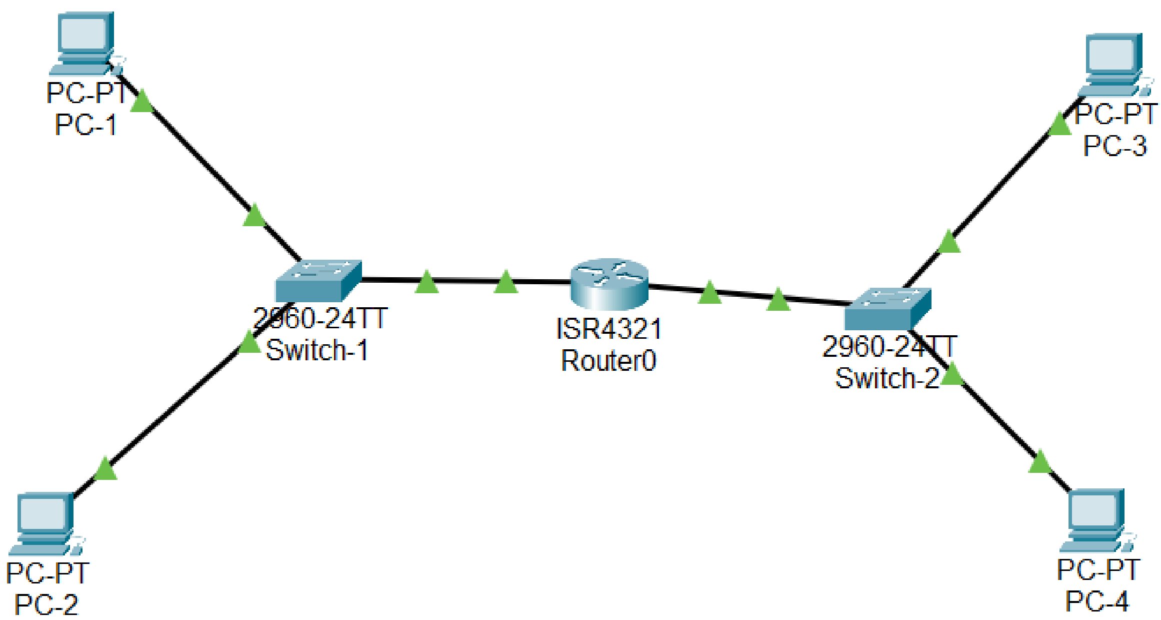


For Router:

- Router interface GigabitEthernet0/0/0 → 192.168.1.1 / 255.255.255.0
- Router interface GigabitEthernet0/0/1 → 192.168.2.1 / 255.255.255.0

Click on Router, go to the CLI tab, and enter the following commands:

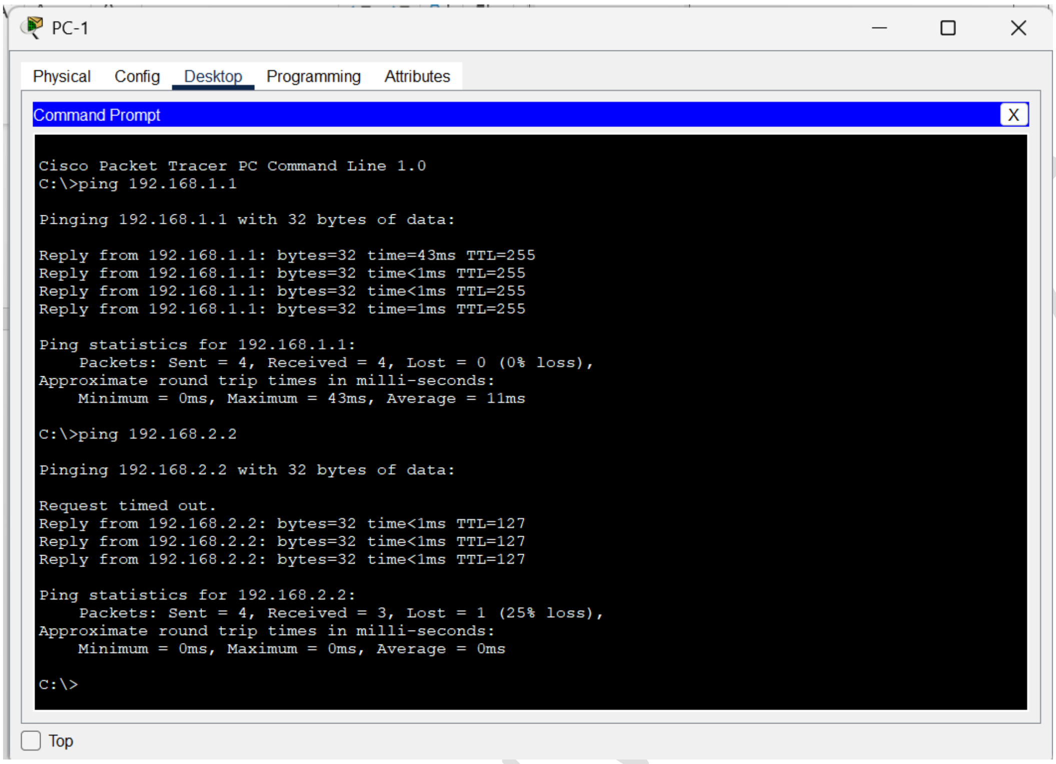
```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   192.168.1.1     YES manual up                    up
GigabitEthernet0/0/1   192.168.2.1     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#
```



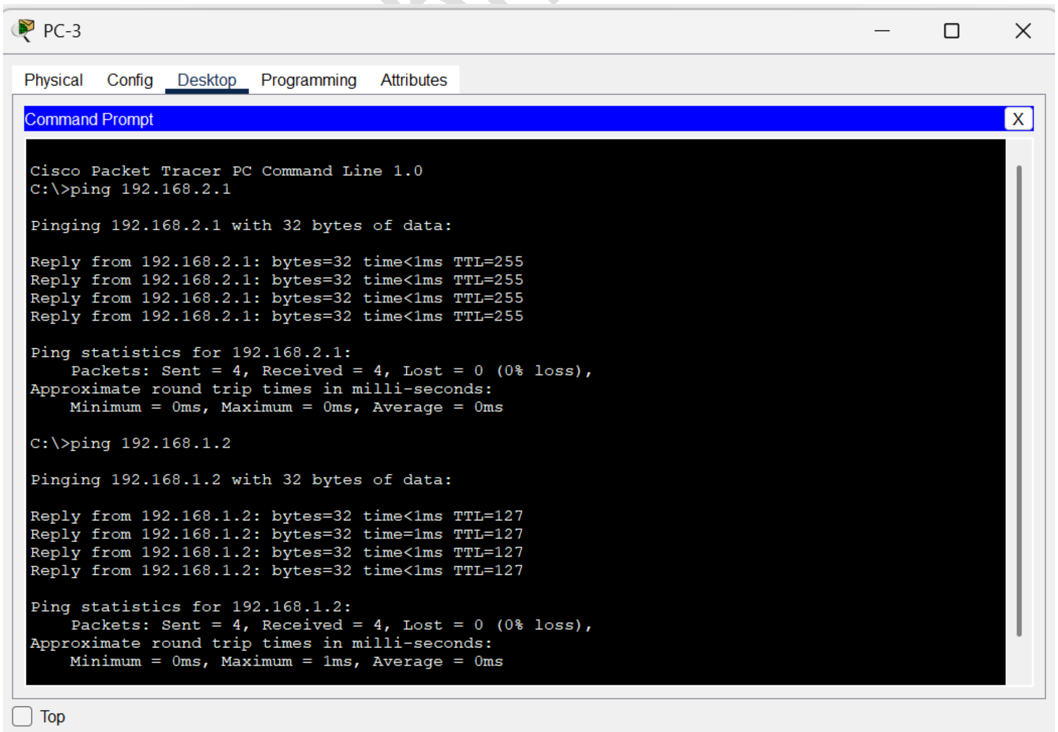
5. Test Connectivity

These test the router and cross-network connectivity.

- On PC1, open the Command Prompt and type:



- On PC3, open the Command Prompt and type:



Successful replies confirm that inter-network communication is functioning correctly.

Result:

Thus, a simple network topology with one router, two switches, and four PCs was successfully designed and configured in Cisco Packet Tracer. Successful ping replies verified proper connectivity between all devices in both networks.